CSCE 411
Linear Programming Practice

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1. You are in charge of a factory making robots out of recycled parts. Each robot requires k different types of components to assemble. In particular, it requires c_i copies of component i for each $1 \leq i \leq k$. There are ℓ different kinds of machines that you can acquire and recycle to get these components. For each $1 \leq i \leq \ell$, each unit of machine i costs d_i dollars and recycling it yields r_{ij} units of component j for each $1 \leq j \leq k$. You have a daily budget of b dollars to spend on buying machines to recycle (you may save money and components between days, however), and you want to maximize the average daily number of robots produced by your factory. Give a polynomial time algorithm to find the maximum possible average daily robot production.

Solution: We give the following linear program:

$$\begin{array}{lrcl} \text{maximize} & R & & \\ \text{subject to} & \sum_{i=1}^{\ell} r_{ij} x_i & \geq c_j R & \forall\, 1 \leq j \leq k \\ & \sum_{i=1}^{\ell} d_i x_i & \leq b & \\ & x_i & \geq 0 & \forall\, 1 \leq i \leq \ell \end{array}$$

Here R represents the average number of robots per day, while x_i represents the average amount of machine i that we buy per day. Note that fractional values for the x_i correspond to buying an extra copy of a machine on some days. e.g. $x_i = 1.5$ means we alternate between buying 1 and 2 copies of machine i. Similarly a fractional value for R represents producing more robots on some days than others.

The first set of constraints encodes the requirement that we need to have enough components to make our robots. For each component, the average amount we acquire from our machine purchases in the day must be enough to produce our average number of robots.

The second constraint encodes the budget restriction, our average daily spending cannot exceed b.

This LP has $\ell + 1$ variables and $k + \ell + 1$ constraints, so we can solve it in $O((k+\ell)^{2.25+o(1)})$ time using the Cohen-Lee-Song algorithm.

2. Convert the LP below to equational form (do not attempt to solve the LP):

$$\begin{array}{ll} \text{minimize} & x_1 + 2x_2 + 3x_3 \\ \text{subject to} & 5x_1 + 3x_2 \leq 7 \\ & 6x_2 + 7x_3 \geq 4 \\ & x_1 \geq 9 \end{array}$$

Solution: We begin by converting the LP to standard form, then convert the standard form to equational form.

First, we change the objective from "minimize $x_1+2x_2+3x_3$" to "maximize $-x_1-2x_2-3x_3$ by multiplying by -1. Similarly, we convert the $\geq$ inequalities into $\leq$ inequalities by multiplying both sides by -1. This gives the new LP:

$$\begin{aligned} \text{maximize} \quad & -x_1 - 2x_2 - 3x_3 \\ \text{subject to} \quad & 5x_1 + 3x_2 \leq 7 \\ & -6x_2 - 7x_3 \leq -4 \\ & -x_1 \leq -9 \end{aligned}$$

Finally, we convert to an LP where every variable is required to be non-negative by replacing x_1, x_2, x_3 with new variables $a_1, a_2, a_3, b_1, b_2, b_3$. We replace each copy of x_i with $a_i - b_i$ (for each $i \in \{1, 2, 3\}$) and require $a_i, b_i \geq 0$. Since any number can be expressed as the difference of two positive numbers, solutions to our new LP correspond to solutions to the original LP. Thus, the standard form version of the original LP is:

$$\begin{aligned} \text{maximize} \quad & -(a_1 - b_1) - 2(a_2 - b_2) - 3(a_3 - b_3) \\ \text{subject to} \quad & 5(a_1 - b_1) + 3(a_2 - b_2) \leq 7 \\ & -6(a_2 - b_2) - 7(a_3 - b_3) \leq -4 \\ & -(a_1 - b_1) \leq -9 \\ & a_1, a_2, a_3, b_1, b_2, b_3 \geq 0 \end{aligned}$$

To convert to equational form, we introduce a slack variable for each of our $\leq$ inequalities, obtaining the following final LP:

$$\begin{aligned} \text{maximize} \quad & -(a_1 - b_1) - 2(a_2 - b_2) - 3(a_3 - b_3) \\ \text{subject to} \quad & 5(a_1 - b_1) + 3(a_2 - b_2) + s_1 = 7 \\ & -6(a_2 - b_2) - 7(a_3 - b_3) + s_2 = -4 \\ & -(a_1 - b_1) + s_3 = -9 \\ & a_1, a_2, a_3, b_1, b_2, b_3, s_1, s_2, s_3 \geq 0 \end{aligned}$$

3. Find the dual of the LP below. If the dual has a feasible solution with an objective value of 50, what can we say about the optimum objective value in the primal? (You do not need to solve the LP)

$$\begin{aligned} \text{maximize} \quad & x_1 + 2x_2 + 3x_4 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 \leq 7 \\ & x_1 + 3x_2 + x_4 \leq 14 \\ & 5x_2 + 4x_3 - 2x_4 \leq 26 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

Solution: The dual is a minimization problem with variables y_1, y_2, y_3 corresponding to the three $\leq$-inequalities in the primal. The right hand side of the dual comes from the objective function of the primal, and the objective function of the dual comes from the right hand side of the primal. We obtain the following LP:

$$\begin{array}{ll} \text{minimize} & 7y_1 + 14y_2 + 26y_3 \\ \text{subject to} & 2y_1 + y_2 \geq 1 \\ & y_1 + 3y_2 + 5y_3 \geq 2 \\ & y_1 + 4y_3 \geq 0 \\ & y_2 - 2y_3 \geq 3 \\ & y_1, y_2, y_3 \geq 0 \end{array}$$

4. Solve this LP with the simplex method, using the pivot rule "pivot to whichever adjacent vertex increases the objective value the most" and starting from the basic feasible solution $x = (0, 3, 0, 1)$

$$\begin{array}{ll} \text{maximize} & x_1 - x_2 + 3x_3 \\ \text{subject to} & x_1 + x_2 + x_3 + x_4 = 4 \\ & x_1 + 2x_2 + 3x_3 + 4x_4 = 10 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{array}$$

Solution: To pivot to an adjacent vertex, we need to bring one of the variables that is currently 0 (x_1 or x_3) into our basic feasible solution. In order to determine which vertex the algorithm pivots to, we need to find both of them and calculate the objective value for each. We note that our starting vertex, $(0, 3, 0, 1)$ has an objective value of -3.

In order to determine how x_2 and x_4 change when we bring a new variable into the basis, we manipulate the equations to isolate the basis variables from each other. Subtracting four times the first equation from the second gives us an equation which involves x_2, but not x_4:

$$-3x_1 - 2x_2 - x_3 = -6. \tag{1}$$

Subtracting twice the first equation from the second gives us an equation which involves x_4, but not x_2.

$$-x_1 + x_3 + 2x_4 = 2. \tag{2}$$

Now we find the vertices adjacent to $(0, 3, 0, 1)$

- Suppose we bring x_1 into the basis (leaving x_3 fixed at 0). From equation (1) above, we have that $x_2 = 3 - \frac{3}{2}x_1$. Thus, as we increase x_1, x_2 will hit 0 at $x_1 = 2$. From equation (2) above, we have that $x_4 = 1 + \frac{1}{2}x_1$, so as we increase x_1, x_4 also increases. Therefore, as we increase x_1, x_2 will hit 0 first (if both x_2 and x_4 were increasing when we increase x_1 then the LP would be unbounded in this direction), so our new vertex is $(2, 0, 0, 2)$, which has an objective value of 2.
- Now suppose we bring x_3 into the basis (leaving x_1 fixed at 0). From equation (1) above, we have that $x_2 = 3 - \frac{1}{2}x_3$. Thus, as we increase x_3, x_2 will hit 0 at $x_3 = 6$. From equation (2) above, we have that $x_4 = 1 - \frac{1}{2}x_3$, so as we increase x_3, x_4 will hit 0 at $x_3 = 2$. Therefore, as we increase x_3, x_4 will hit 0 first, so our new vertex is $(0, 2, 2, 0)$, which has an objective value of 4.

Comparing the two adjacent vertices we found, we conclude that the one that the algorithm will pivot to is $(0, 2, 2, 0)$, since this provides the largest increase in the objective value.

Rewriting our LP to isolate our new basis of x_2, x_3 we get the equations:

$$\begin{aligned} 2x_1 \ + x_2 \quad\quad - x_4 \ &= 2 \\ -\ x_1 \quad\quad + x_3 + 2x_4 &= 2 \end{aligned}$$

For our next pivot, we have only one candidate. Bringing x_4 back into the basis would return us to the vertex we started at, so we consider bringing x_1 into the basis. From our equations we have that $x_2 = 2 - 2x_1$ and $x_3 = 2 + x_1$, so as we increase x_1, x_2 is the only variable that decreases and it hits 0 at $x_1 = 1$. Thus, our new vertex is $(1, 0, 3, 0)$, with an objective value of 10. This is larger that our current objective value of 4, so we pivot to this vertex.

Rewriting our LP to isolate our new basis of x_1, x_3 we get the equations:

$$\begin{aligned} x_1 + 0.5x_2 \quad\quad - 0.5x_4 &= 1 \\ + 0.5x_2 + x_3 + 1.5x_4 &= 3 \end{aligned}$$

Again we have one candidate for the next pivot. Consider bringing x_4 into the basis. We have $x_1 = 1 + 0.5x_4$ and $x_3 = 3 - 1.5x_4$, so as we increase x_4, x_3 hits 0 at $x_4 = 2$, giving us the vertex $(2, 0, 0, 2)$, with objective value 2. This is worse than the vertex we are currently at, so we do not pivot, and conclude that $(1, 0, 3, 0)$ is an optimal solution.